VDS4E BRIDGE ONE — SCHEMATIC OVERVIEW

USB-C (any source, 5-24 V tolerated) -> PTC + TVS -> AP63203 buck -> +3V3 for the whole board.
ESP32-S3: USB to the PC, I2C0/I2C1 + UART target ports, OLED, FPGA configuration and control.
iCE40UP5K: target SPI engine (same-CS decode) + 8-channel GPIO bank + QSPI PSRAM flash image.
All target signals pass level translators (A = +3V3, B = VTREF) and ESD; isolated by default.
TARGET I/O: Raspberry Pi 40-pin header + labelled direct header with 5V_OUT/3V3_OUT accessories.
Generated from cad/schematic/design.py; verify with tools/verify_netlist.py after every change.

USB-C universal power input

A3 — 8 parts

File: power_in.kicad_sch

3V3 buck (3.8-32 V in)

A3 — 10 parts

File: buck.kicad_sch

ESP32-S3 control plane

A3 — 25 parts

File: mcu.kicad_sch

Aux power outputs (J10 5V_OUT / 3V3_OUT)

A3 — 21 parts

File: aux_out.kicad_sch

Status LEDs

A3 — 2 parts

File: status.kicad_sch

iCE40UP5K SPI/GPIO bus engine

A3 — 41 parts

File: fpga.kicad_sch

Target isolation logic + VTREF sense

A3 — 34 parts

File: gating.kicad_sch

Target level translators and I2C repeaters

A3 — 57 parts

File: translators.kicad_sch

TARGET I/O (back edge)

A3 — 7 parts

File: target_io.kicad_sch

Status OLED 0.96in 128x64 (SSD1315, I2C 0x3C)

A3 — 10 parts

File: oled.kicad_sch

Generated from cad/schematic/design.py — do not edit the .kicad_sch by hand
VDS4E

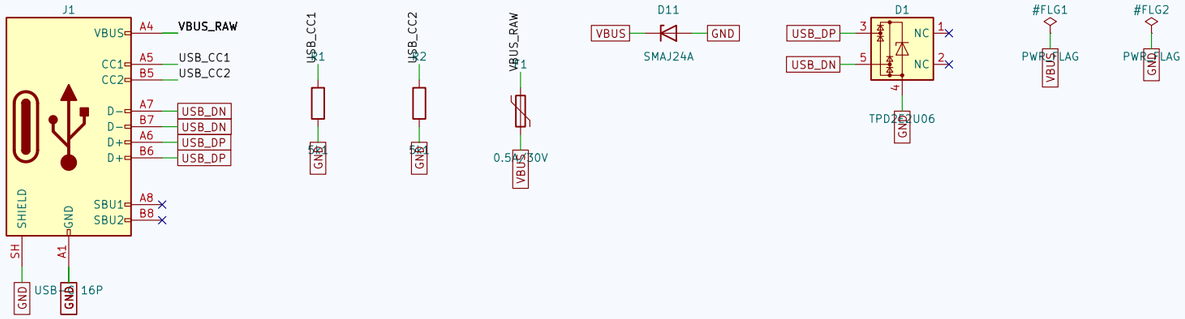
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File: bridge-one.kicad_sch

Title: VDS4E Bridge One — Overview

Size: A3	Date: 2026-10-14	**Rev: F-sch0.1**
KiCad E.D.A. 10.0.6		Id: 1/11

USB-C UNIVERSAL POWER INPUT

- *Non-PD sink: 5.1k Rd on CC1/CC2 -> compliant sources give 5 V only.*
- *Survives non-compliant 9-24 V on VBUS: 30 V PTC + SMAJ24A clamp, buck rated 32 V.*
- *USB D+/D- ESD has no VBUS reference pin (TPD2E2U06), so VBUS faults cannot reach it.*
- *Ref: research/power-input-universal.md*



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VDS4E

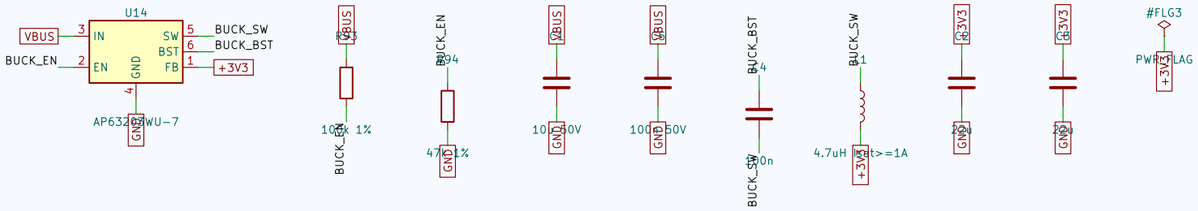
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Title: VDS4E Bridge One — USB-C universal power input

Size: A3	Date: 2026-10-14	**Rev: F-sch0.1**
KiCad E.D.A. 10.0.6		Id: 2/11

3V3 BUCK (3.8–32 V IN)

- *AP63203: fixed 3.3 V, 2 A, 1.1 MHz, PFM at light load.*
- *EN from 100k/47k divider: EN <= 12.4 V at the 38.9 V TVS clamp (EN abs max 35 V); starts at ~3.7 V VIN.*
- *Board load ~100 mA typ, ~225 mA worst with FPGA; 3V3_OUT adds up to 300 mA.*
- *Cin rated 50 V (VBUS may sit at 24 V below the SMAJ24A clamp); C1/C5 at IN–GND, L1/Cout close to SW.*



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VDS4E

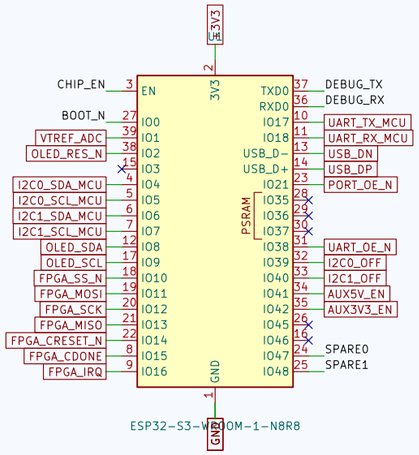
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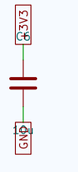
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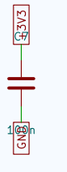
Size: A3	Date: 2026-10-14	**Rev: F-sch0.1**
KiCad E.D.A. 10.0.6		Id: 3/11

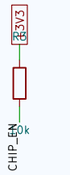
ESP32-S3 CONTROL PLANE

- *Native USB (GPIO19/20) = programming, serial/JTAG and PC link; no USB-UART chip.*
- *Both hardware I2C controllers serve the target as slaves; OLED uses software I2C.*
- *Reset defaults: target ports isolated, FPGA held in reset, aux outputs off.*
- *Straps GPIO3/45/46 unconnected; GPIO35-37 belong to the octal PSRAM.*
- *Ref: docs/pin-allocation.md*

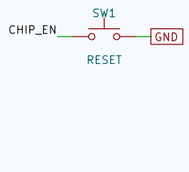


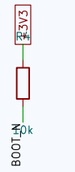


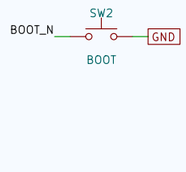


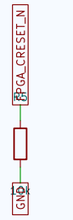


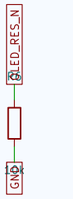


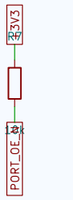


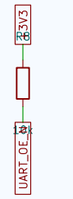


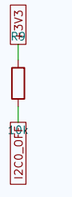


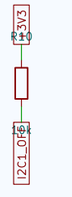


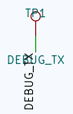


















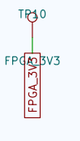














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VDS4E

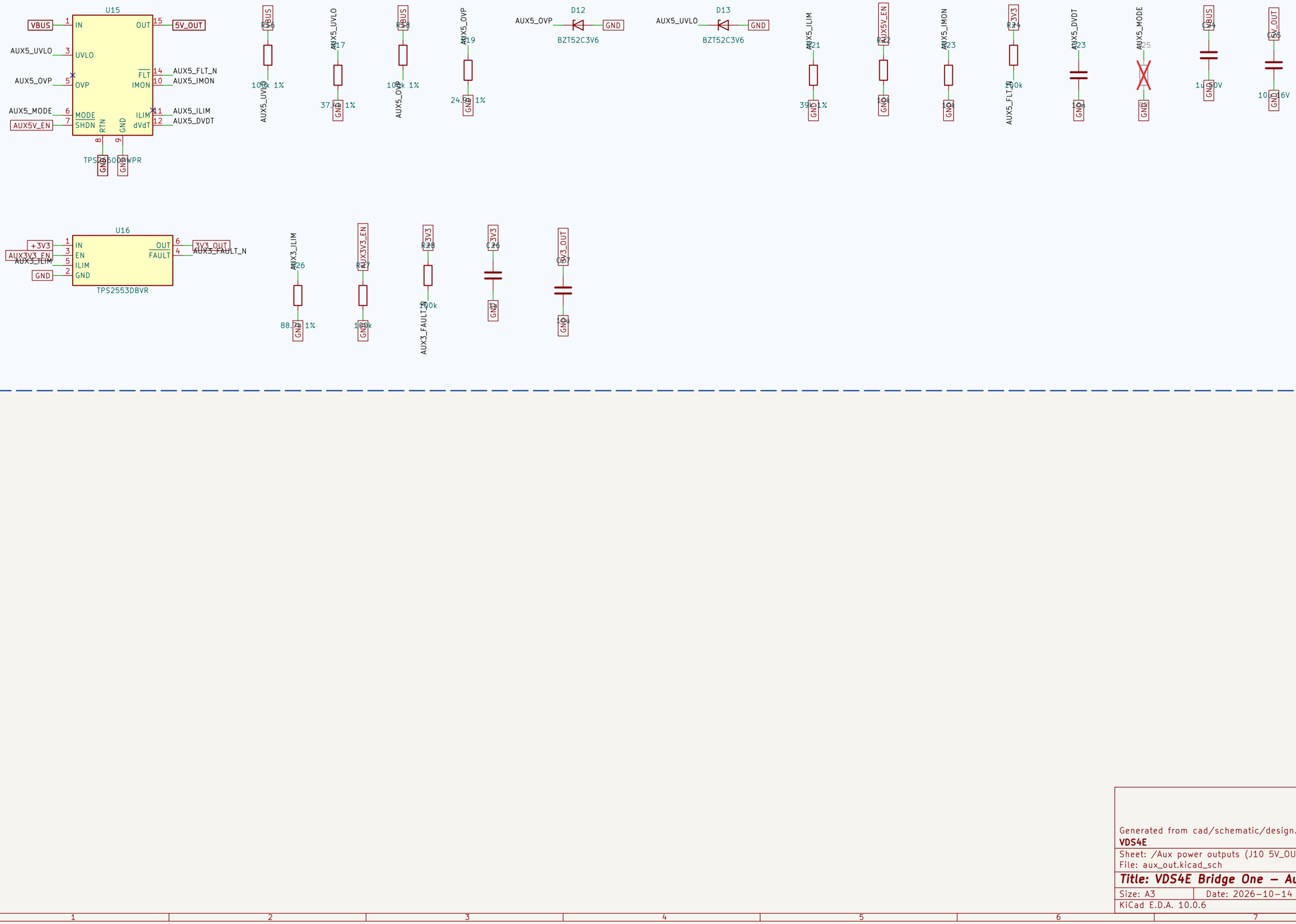
Sheet: /ESP32-S3 control plane/
File: mcu.kicad_sch

Title: VDS4E Bridge One — ESP32-S3 control plane

Size: A3	Date: 2026-10-14	**Rev: F-sch0.1**
KiCad E.D.A. 10.0.6		Id: 4/11

AUX POWER OUTPUTS (J10 5V_OUT / 3V3_OUT)

- *Off at reset (pull-downs on SHDN/EN); firmware enables per user request.*
- *5V_OUT: TPS26600 opens above ~6.0 V VBUS (OVP 100k/24.9k), UVLO ~4.4 V, ILIM 0.3 A (39k).*
- *D12/D13 3.6 V zeners keep OVP/UVLO pins below their 5 V abs max at 24 V and during TVS clamping.*
- *3V3_OUT: TPS2553 limited to ~300 mA (88.7k), blocks reverse current from the user device.*
- *Never tied to VREF or the Raspberry Pi 5 V / 3.3 V pins (no back-powering).*
- *Ref: research/aux-power-outputs.md*



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VDS4E

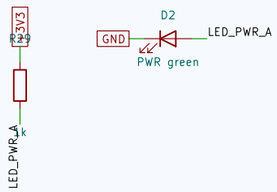
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File: aux_out.kicad_sch

Title: VDS4E Bridge One — Aux power outputs (J10 5V_OUT / 3V3_OUT)

Size: A3	Date: 2026-10-14	**Rev: F-sch0.1**
KiCad E.D.A. 10.0.6		Id: 5/11

STATUS LEDS

- *PWR is hard-wired to +3V3. LINK / VTREF / FPGA LEDs sink into iCE40 RGB0-2 (FPGA group).*



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VDS4E

Sheet: /Status LEDs/
File: status.kicad_sch

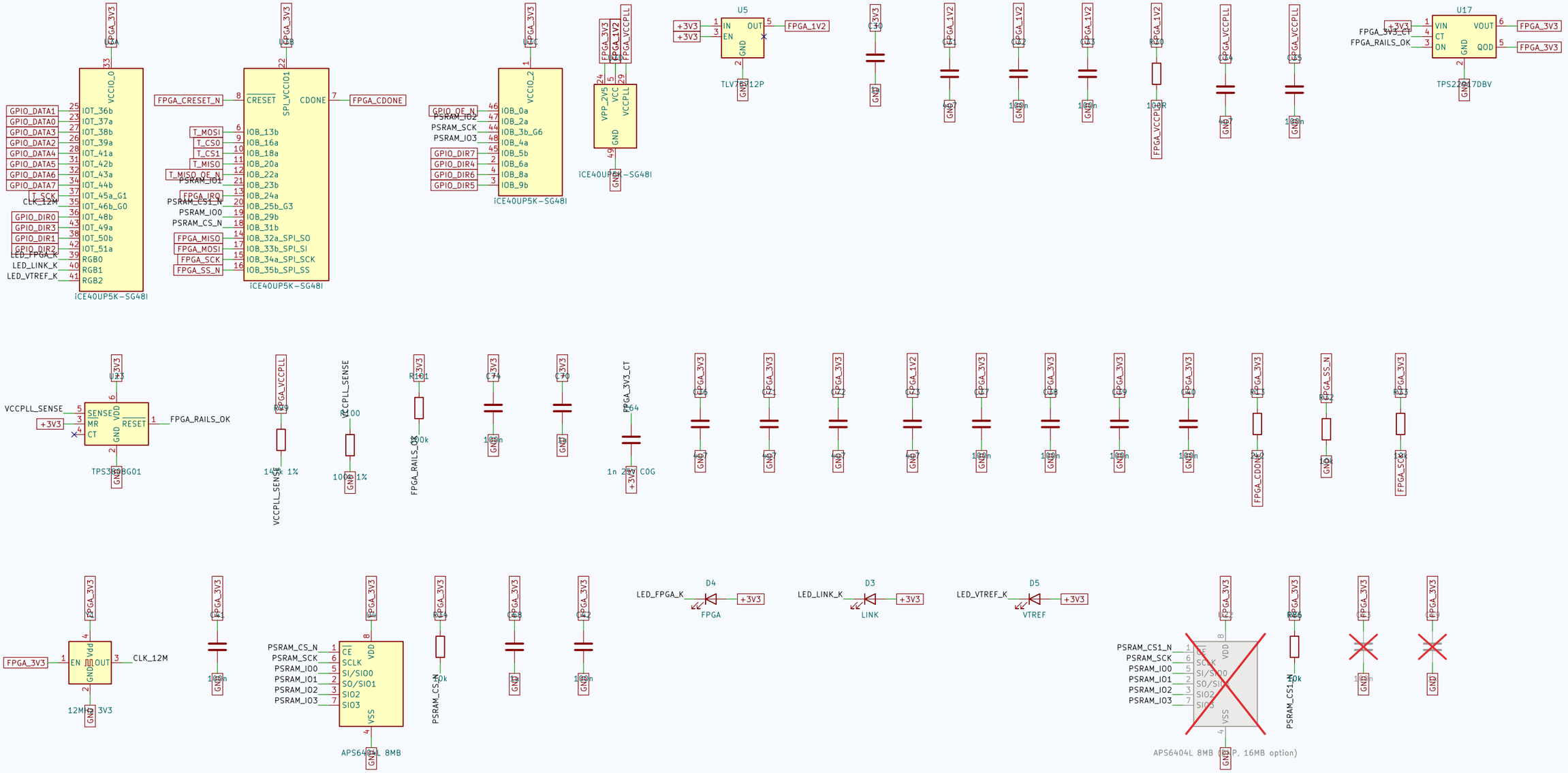
Title: VDS4E Bridge One — Status LEDs

Size: A3	Date: 2026-10-14	**Rev: F-sch0.1**
KiCad E.D.A. 10.0.6		Id: 6/11

ICE40UP5K SPI/GPIO BUS ENGINE

- *Configured by the S3 over SPI (slave mode, no config flash); same 4 wires become the data link.*
- *Sequence: TPS3808G01 releases FPGA_RAILS_OK 20 ms after VCCPLL (lags core) > 1.00 V (147k/100k); then TPS22917 ramps FPGA_3V3 ~1.8 V/ms.*
- *Pins per research/ice40-pin-plan.md: CLK on G0/PLL pin 35, target SCK on G1 pin 37.*

- *RGB0-2 current sinks drive the FPGA / LINK / VTREF LEDs (anodes on +3V3, no series R).*
- *PSRAM: 8 MB fitted, second 8 MB footprint DNP (pin 20). 32 MB needs 2 more CS pins (open).*
- *READ 0x03 arbitrary address ~1 MHz; FAST_READ 0x0B ~20 MHz (calculated, research/spi-backing-memory.md).*



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VDS4E

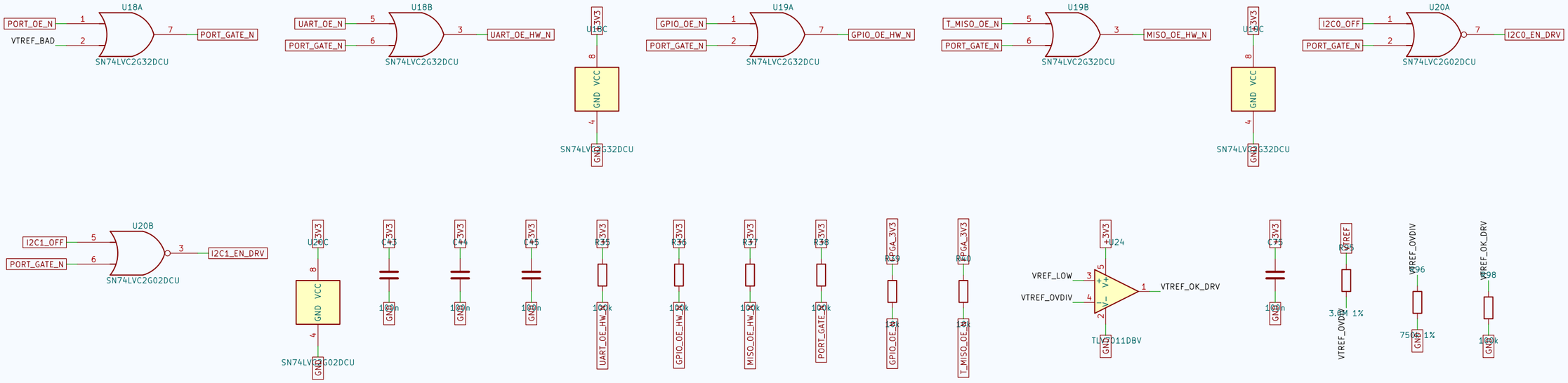
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File: fpga.kicad_sch

Title: VDS4E Bridge One — iCE40UP5K SPI/GPIO bus engine

Size: A3	Date: 2026-10-14	**Rev: F-sch0.1**
KiCad E.D.A. 10.0.6		Id: 7/11

TARGET ISOLATION LOGIC + VTREF SENSE

- *Global wins: PORT_GATE_N = PORT_OE_N OR VTREF_BAD; every port enable is gated by it in hardware.*
- *Reset default = isolated (pull-ups on *_OE_N / *_OFF, FPGA outputs Hi-Z until configured).*
- *VTREF_BAD when VTREF < ~1.5 V (TLV7011 vs 0.76 V ref, own unfiltered 47k/47k divider: fast); runs from +3V3.*
- *VTREF is an input reference only: 470k/470k divider keeps backfeed ~3 uA.*
- *OVP: Q1 P-FET passes VTREF only while U24 sees VTREF < 3.81 V (3.0M/750k vs VREF_LOW) and +3V3 is up; else VTREF_IO is cut.*
- *Q2 (60 V) takes the fault voltage instead of the comparator output. Ref: research/vtref-ovp-and-fpga-supervisor.md*
- *Ref: research/target-interface-spec.md, research/pinouts-and-gates.md*



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VDS4E

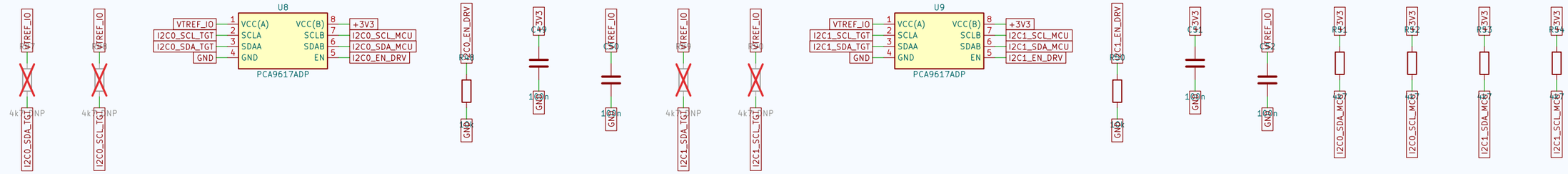
Sheet: /Target isolation logic + VTREF sense/
File: gating.kicad_sch

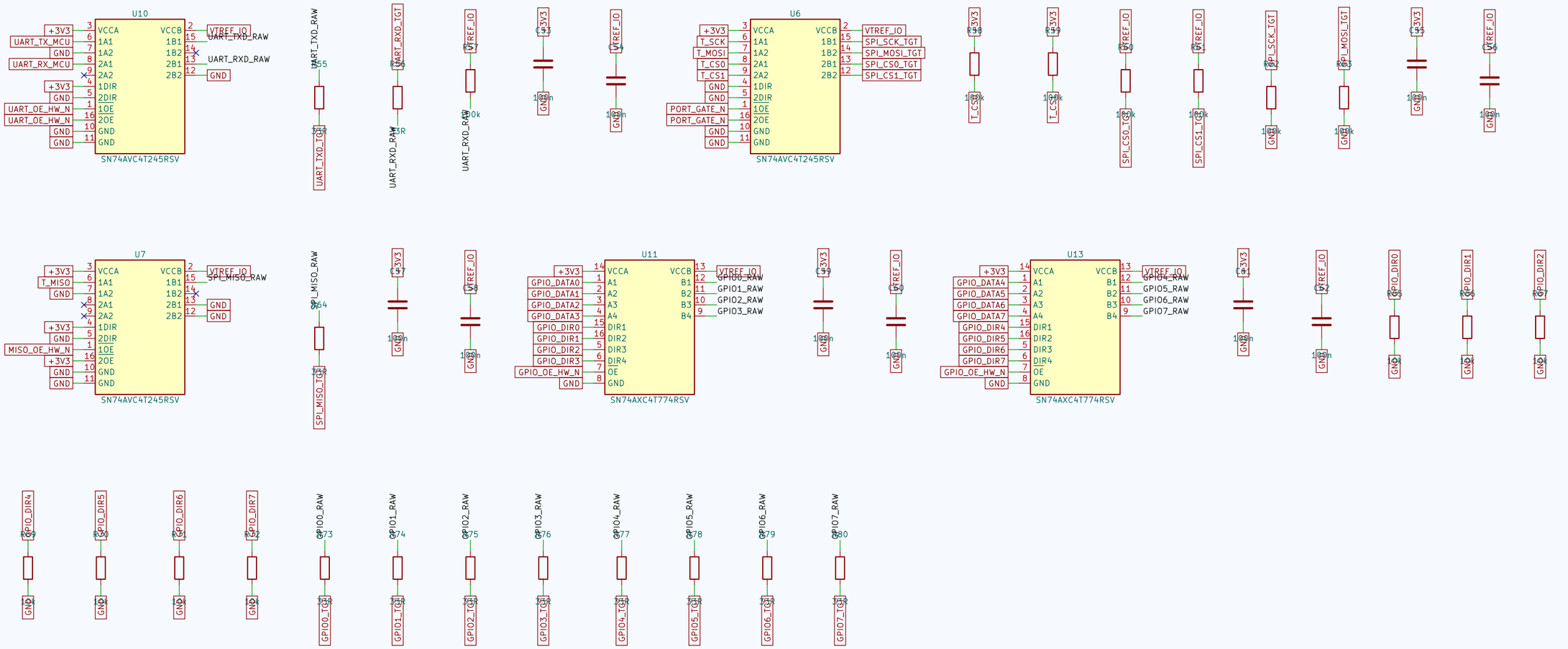
Title: VDS4E Bridge One — Target isolation logic + VTREF sense

Size: A3	Date: 2026-10-14	**Rev: F-sch0.1**
KiCad E.D.A. 10.0.6		Id: 8/11

TARGET LEVEL TRANSLATORS AND I2C REPEATERS

- *AVC/AXC: VCCA = +3V3, VCCB = VTREF_IO (1.65–3.6 V); DIR/OE referenced to VCCA; Ioff isolates when unpowered.*
- *I2C: PCA9617A repeaters, port A = target (VTREF_IO 0.8–5.5 V), port B = MCU 3.3 V; equal rails supported.*
- *EN = gated I2Cn_EN_DRV (10k pull-down); no bias path from +3V3 into VTREF. B side has <=0.6 V static offset.*
- *BENCH: ESP32 open-drain low <= 0.4 V under load (B-side requirement); 1.65 V targets have ~12 mV VIL margin.*
- *UART: bank 1 bridge->target, bank 2 target->bridge, fixed. SPI: U6 target->FPGA, U7 MISO only.*
- *GPIO: per-bit DIR from FPGA (10k pull-down = B->A input default), common OE gated.*
- *MCU-side I2C pull-ups 4k7 fitted; target-side 4k7 to VTREF_IO DNP (R87–R90): targets usually have their own.*

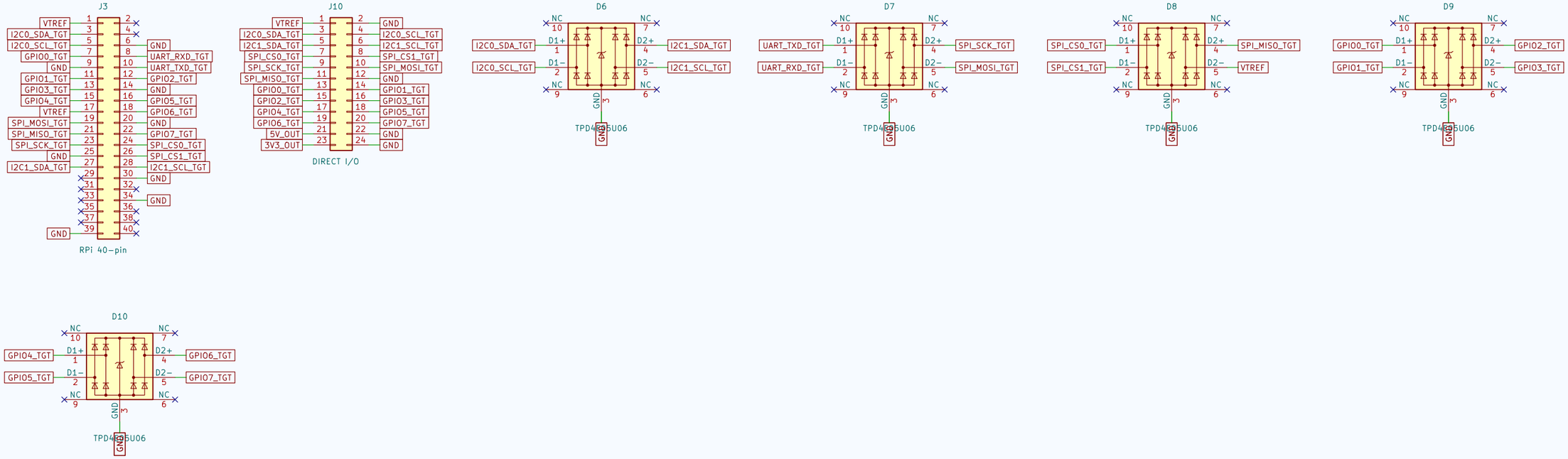




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VDS4E
Sheet: /Target level translators and I2C repeaters/
File: translators.kicad_sch
Title: VDS4E Bridge One — Target level translators and I2C repeaters
Size: A3 | Date: 2026-10-14 | **Rev: F-sch0.1**
KiCad E.D.A. 10.0.6 | Id: 9/11

TARGET I/O (BACK EDGE)

- *J3 = Raspberry Pi 40-pin numbering (ribbon cable); J10 = labelled direct-plug header. Same nets.*
- *Pi 5 V pins (2/4) are NOT connected; VTREF (pins 1/17) is an input reference only.*
- *J10 21/23 = 5V_OUT / 3V3_OUT accessory supplies (aux_out group). Connect one target at a time.*
- *TPD4E05U06 flow-through: route each line straight under the part (NC pads are pass-through).*
- *Ref: docs/pi-header-pinout.md*



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VDS4E

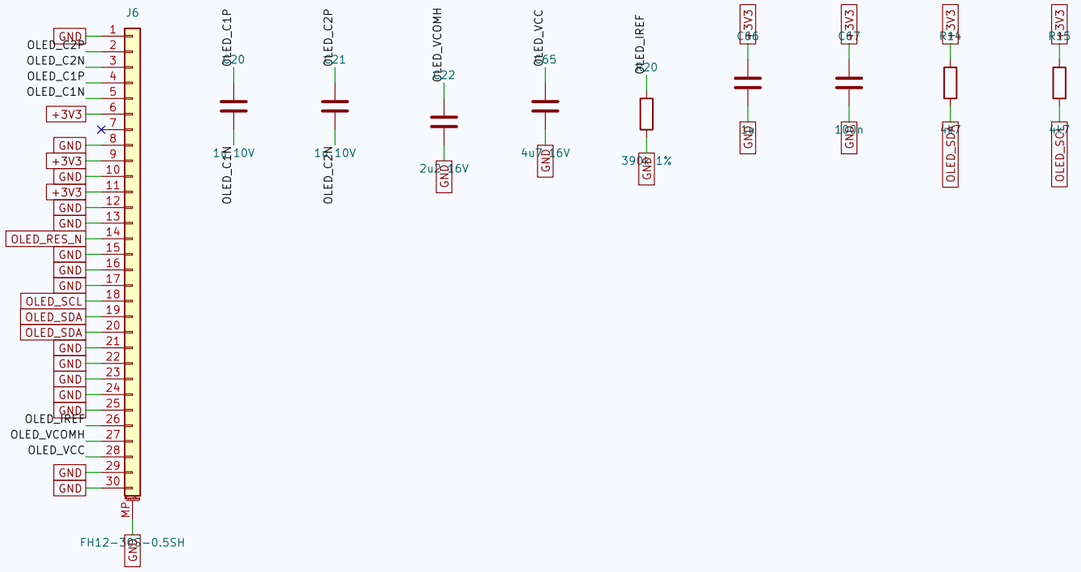
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File: target_io.kicad_sch

Title: VDS4E Bridge One — TARGET I/O (back edge)

Size: A3	Date: 2026-10-14	**Rev: F-sch0.1**
KiCad E.D.A. 10.0.6		Id: 10/11

STATUS OLED 0.96IN 128X64 (SSD1315, I2C 0X3C)

- *Panel: Wisevision X096-2864KSWPG17-C30 (30-pin 0.5 mm ZIF tail); 2nd source ER-OLED0.96-1.3W.*
- *I2C mode: BS0=0, BS1=1, BS2=0, CS#=0, D/C#(SA0)=0 -> address 0x3C; D1+D2 tied = SDA.*
- *Internal charge pump from +3V3 (VBAT): 2x 1 uF flying caps, VCC 4.7 uF, VCOMH 2.2 uF.*
- *IREF 390k ~ 12.5 uA (560k = dimmer). Software I2C master on GPIO8/9; reset held low at boot.*
- *REVIEW: FPC contact side / fold direction decides whether panel pin 1 meets connector pin 1.*
- *Ref: research/oled-panel.md*



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VDS4E

Sheet: /Status OLED 0.96in 128x64 (SSD1315, I2C 0x3C)/
File: oled.kicad_sch

Title: VDS4E Bridge One — Status OLED 0.96in 128x64 (SSD1315, I2C 0x3C)

Size: A3	Date: 2026-10-14	**Rev: F-sch0.1**
KiCad E.D.A. 10.0.6		Id: 11/11